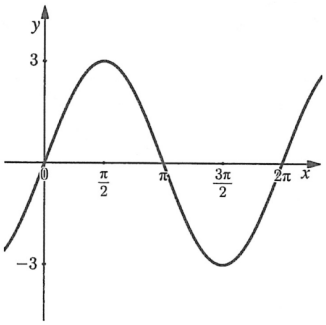
Graph Sketching: sin(x), cos(x), tan(x) in Radians

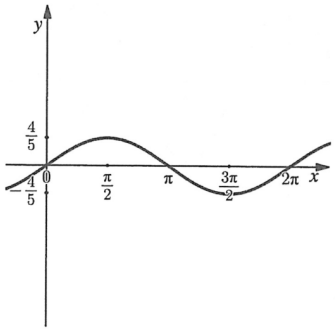
Questions 1–25

Question 1. Find the equation of each of the following graphs in the form $y = a\sin x$

a)

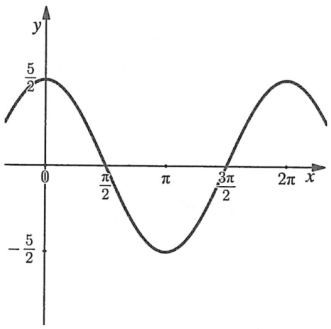


b)

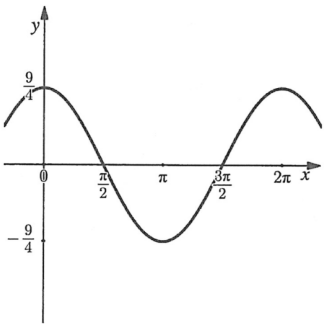


Question 2. Find the equation of each of the following graphs in the form $y = a\cos x$

a)



b)



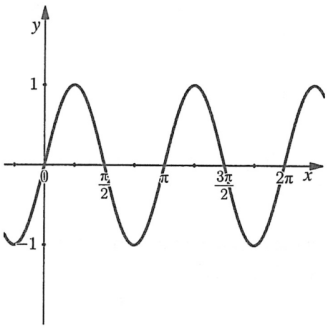
Question 3. Sketch the following graphs for $0 \leq x \leq 2\pi$:

a) $y = \frac{1}{3}\sin x$

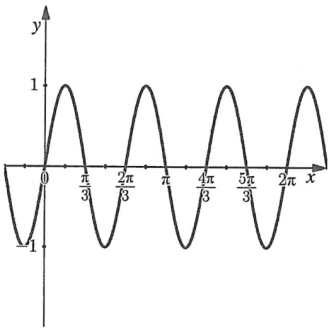
b) $y = \frac{8}{7}\cos x$

Question 4. Find the equation of each of the following graphs in the form $y = \sin ax$

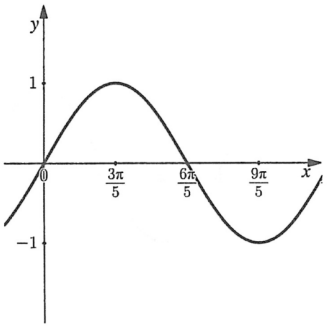
a)



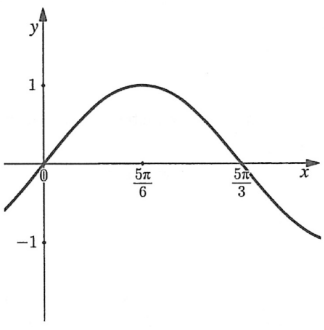
b)



c)



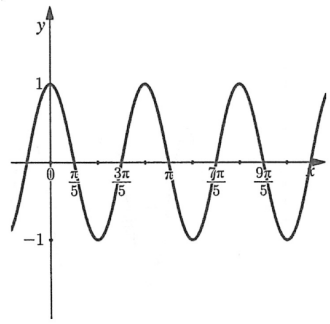
d)



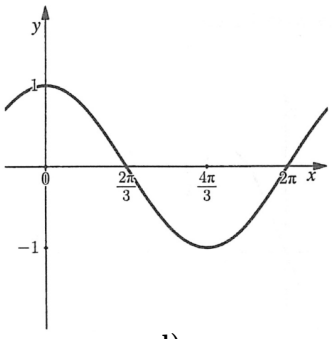
Question 5. Sketch the graph of $y = -4\sin x$ for $0 \leq x \leq 2\pi$

Question 6. Find the equation of each of the following graphs in the form $y = \cos ax$

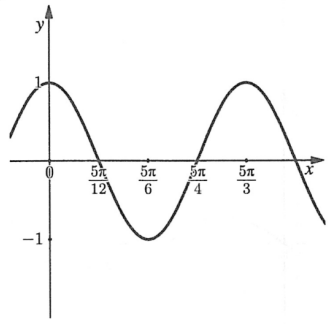
a)



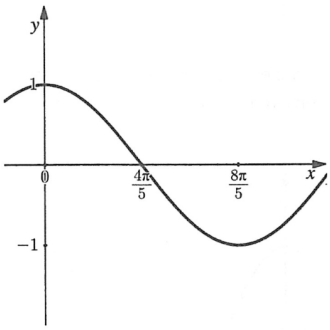
b)



c)



d)

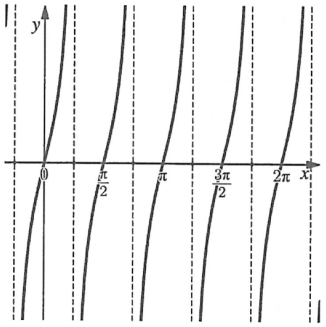


Question 7. Sketch the graph of $y = -\frac{3}{2}\cos x$ for $0 \leq x \leq 2\pi$

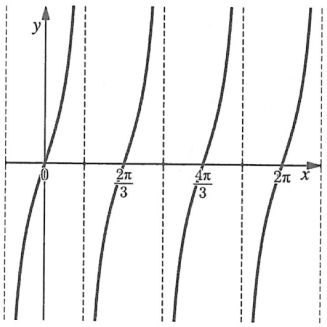
Question 8. Sketch the graph of $y = \sin(-\frac{2}{3}x)$ for $0 \leq x \leq 2\pi$

Question 9. Find the equation of each of the following graphs in the form $y = \tan ax$

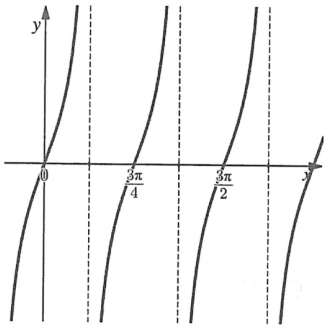
a)



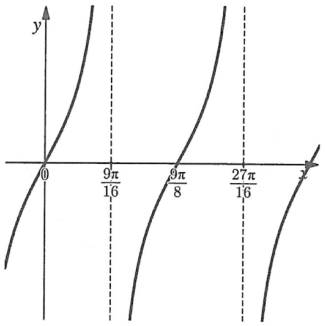
b)



c)



d)



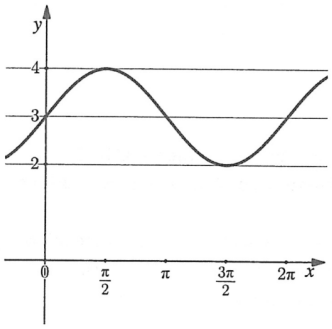
Question 10. Sketch the following graphs for $0° \leq x \leq 2\pi$:

a) $y = \tan \frac{2}{5}x$

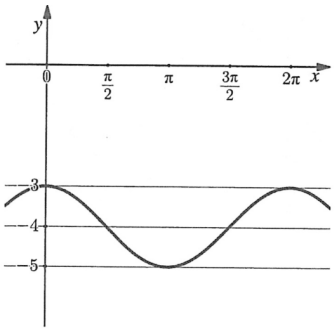
b) $y = \tan \frac{10}{9}x$

Question 11. Find the equation of each of the following graphs in either the form $y = a + \sin x$ or the form $y = a + \cos x$

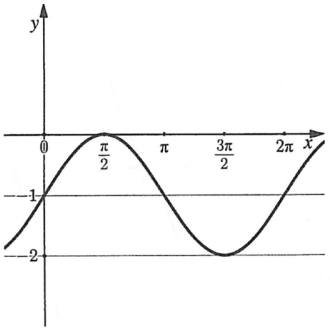
a)



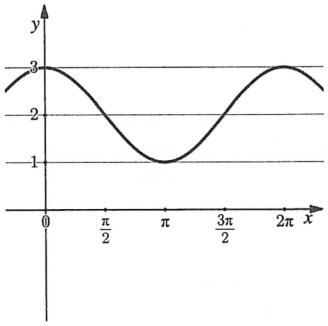
b)



c)

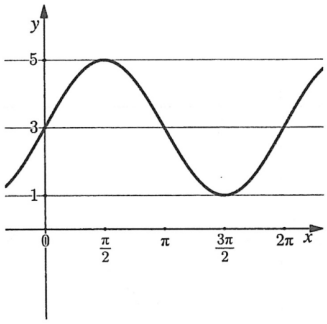


d)

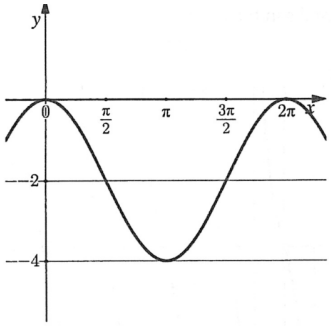


Question 12. Prerequisite knowledge for Q12–Q13: Combinations of Graph Transformations
Find the equation of each of the following graphs in the form $y = a + b\sin x$ or the form $y = a + b\cos x$

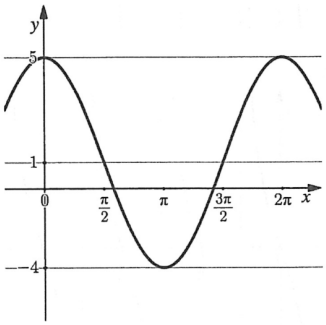
a)



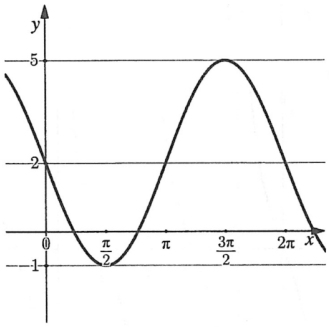
b)



c)



d)



Question 13. Sketch the following graphs for $0 \leq x \leq 2\pi$:

a) $y = 1 + 3\sin x$

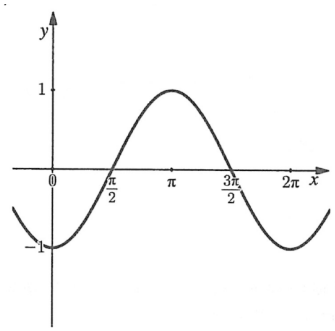
b) $y = -2 + 3\cos x$

c) $y = \frac{3}{2} - 4\sin x$

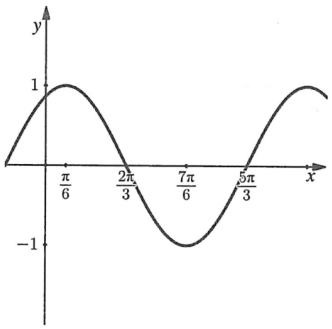
d) $y = -\frac{2}{3} - 2\cos x$

Question 14. Find the equation of each of the following graphs in the form $y = \sin(x + a)$ where $-\pi < a < \pi$

a)

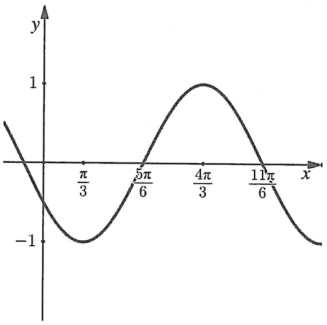


b)

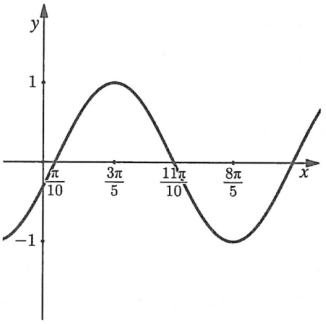


Question 15. Find the equation of each of the following graphs in the form $y = \cos(x + a)$ where $-\pi < a < \pi$

a)

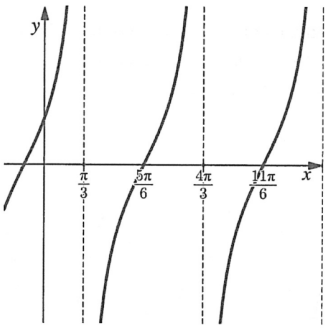


b)

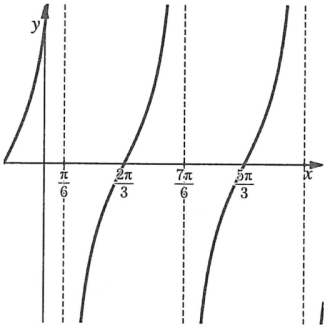


Question 16. Find the equation of each of the following graphs in the form $y = \tan(x + a)$ where $-\frac{\pi}{2} < a < \frac{\pi}{2}$

a)



b)



Question 17. It is given that k is a constant where $k > 1$. Sketch the following graphs for $0 \leq x \leq 2\pi$:

a) $y = -k + \sin x$

b) $y = k + \cos x$

Question 18. Sketch the following graphs for $0 \leq x \leq 2\pi$:

a) $y = \sin(x - \frac{\pi}{3})$

b) $y = \cos(x + \frac{\pi}{3})$

c) $y = \tan(x - \frac{\pi}{5})$

d) $y = \sin(x + \frac{4\pi}{5})$

e) $y = \cos(x - \frac{\pi}{6})$

f) $y = \tan(x + \frac{3\pi}{8})$

Question 19. It is given that k is a constant where $0 < k < \frac{\pi}{2}$. Sketch the following graphs for $0 \leq x \leq 2\pi$:

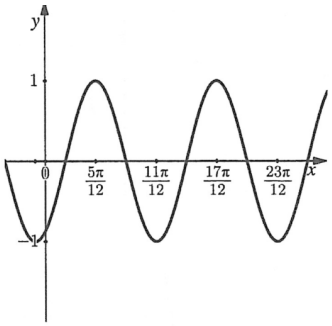
a) $y = \sin(x - k)$

b) $y = \tan(x + k)$

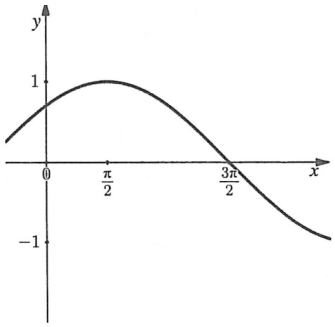
Question 20. Prerequisite knowledge for Q20–Q25: Combinations of Graph Transformations

Find the equation of each of the following graphs in the form $y = \sin(ax + b)$ where $-\pi < b < \pi$

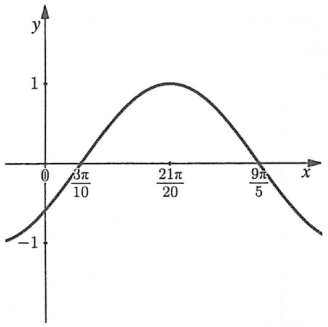
a)



b)

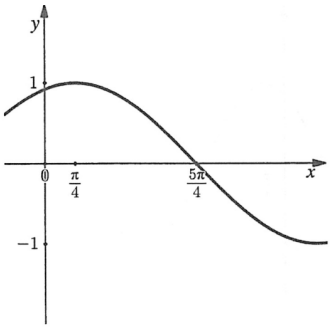


c)

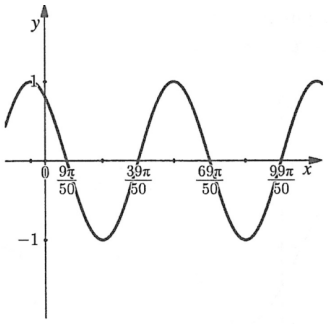


Question 21. Find the equation of each of the following graphs in the form $y = \cos(ax + b)$ where $-\pi < b < \pi$

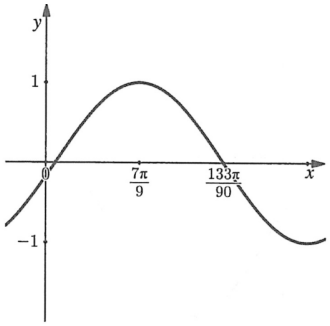
a)



b)

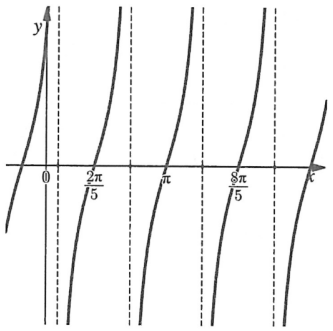


c)

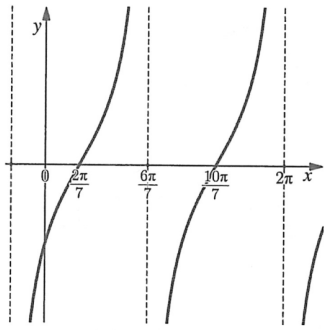


Question 22. Find the equation of each of the following graphs in the form $y = \tan(ax + b)$ where $-\frac{\pi}{2} < b < \frac{\pi}{2}$

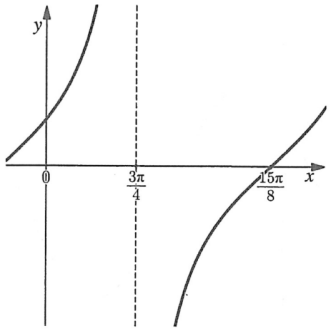
a)



b)

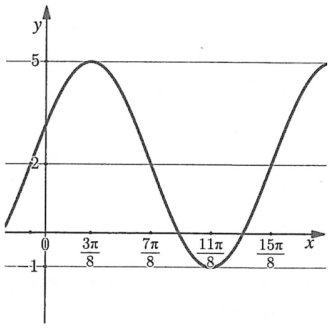


c)

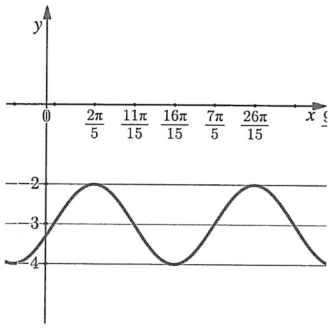


Question 23. Find the equation of each of the following graphs in the form $y = a + b\sin(cx + d)$ where $-\pi < d < \pi$

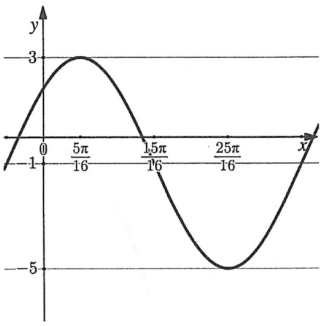
a)



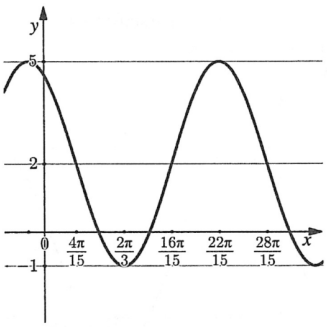
b)



c)

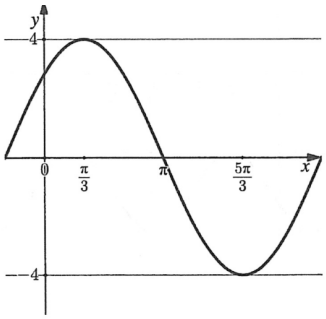


d)

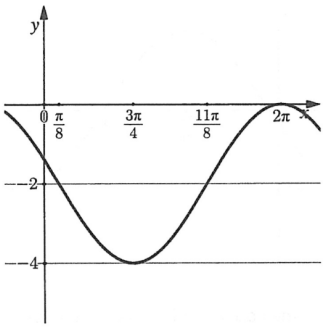


Question 24. Find the equation of each of the following graphs in the form $y = a + b\cos(cx + d)$ where $-\pi < c < \pi$

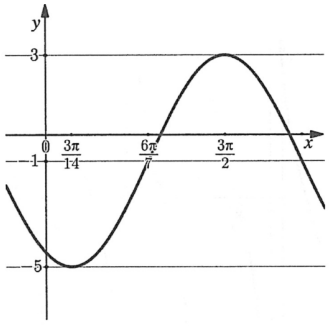
a)



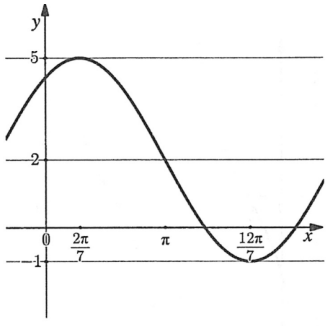
b)



c)



d)



Question 25. It is given that a and b are positive constants. Sketch the following graphs for $0 \leq x < 2\pi$:

a) $y = a + b\cos x$ where $a > b$

b) $y = a + b\cos x$ where $a = b$

c) $y = a + b\cos x$ where $a < b$